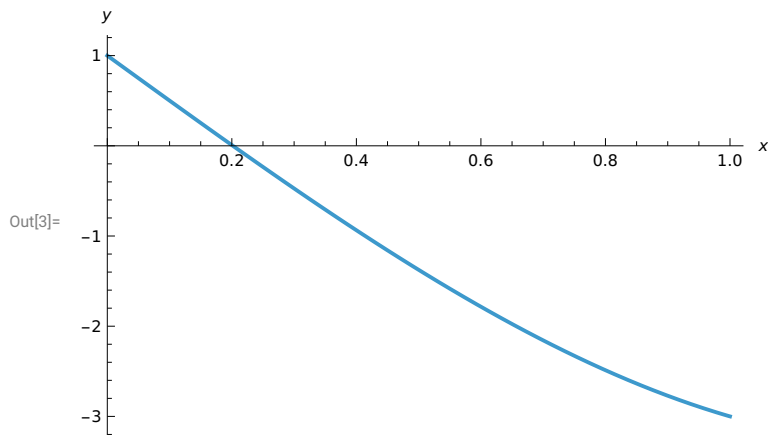


```

In[1]:= Bisection[a0_, b0_, k_] :=
Module[{
a = N[a0];
b = N[b0];
m = (a + b) / 2;
i = 0;
output = {{i, a, b, m, f[a], f[b], f[m], Sign[f[a]*f[m]]}};
While[i < k, If[Sign[f[b]]*Sign[f[m]] < 0, a = m, b = m];
m = (a + b) / 2;
i = i + 1;
output = Append[output, {i, a, b, m, f[a], f[b], f[m], Sign[f[a]*f[m]]}];
Print[
NumberForm[
TableForm[output,
TableHeadings ->
{None, {"i", "a0", "a1", "a2", "f[a0]", "f[a1]", "f[a2]", "Sign[f[a0]*f[a2]]"}}, 16]];
Print["m=", NumberForm[m, 16]];]
f[x_] := x^3 - 5*x + 1
Plot[f[x], {x, 0, 1}, PlotStyle -> {Thick}, AxesLabel -> {x, y}]
Print["y=f[x]=", f[x]];
Bisection[0, 1, 8]

```



$$y=f[x]=1-5x+x^3$$

i	a0	a1	a2	f[a0]	f[a1]
0	0.	1.	0.5	1.	-3.
1	0.	0.5	0.25	1.	-1.375
2	0.	0.25	0.125	1.	-0.234375
3	0.125	0.25	0.1875	0.376953125	-0.234375
4	0.1875	0.25	0.21875	0.069091796875	-0.234375
5	0.1875	0.21875	0.203125	0.069091796875	-0.083282470703125
6	0.1875	0.203125	0.1953125	0.069091796875	-0.007244110107421875
7	0.1953125	0.203125	0.19921875	0.03088808059692383	-0.007244110107421875
8	0.19921875	0.203125	0.201171875	0.01181286573410034	-0.007244110107421875

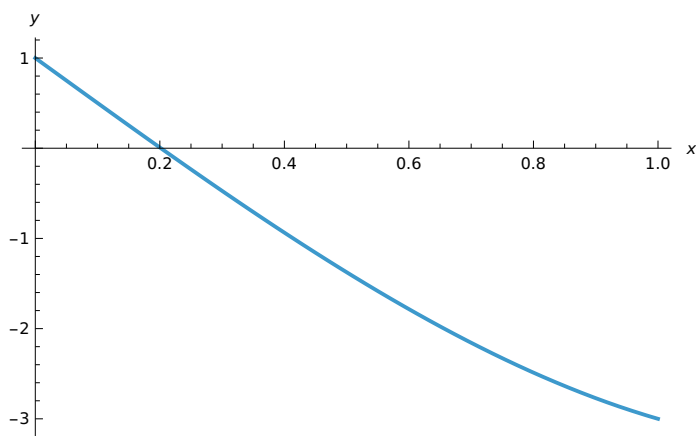
m=0.201171875

```

In[27]:= Secant[a0_, b0_, n_] :=
Module[{i,
a = N[a0];
b = N[b0];
c = (a * f[b] - b * f[a]) / (f[b] - f[a]);
i = 0;
Output = {{i, a, b, c, f[c]}};
While[i < n, a = b; b = c;
c = (a * f[b] - b * f[a]) / (f[b] - f[a]);
i = i + 1;
Output = Append[Output, {i, a, b, c, f[c]}];];
Print[NumberForm[
TableForm[Output, TableHeadings -> {None, {"i", "a", "b", "c", "f[c]"}}, 16]];
Print["c=", NumberForm[c, 16]];]
f[x_] := x^3 - 5 x + 1;
Plot[f[x], {x, 0, 1}, PlotStyle -> {Thick}, AxesLabel -> {x, y}]
Print["y=f[x]=", f[x]];
Secant[0, 1, 6]

```

Out[29]=



$$y=f[x]=1-5x+x^3$$

i	a	b	c	f[c]
0	0.	1.	0.25	-0.234375
1	1.	0.25	0.1864406779661017	0.0742773117017806
2	0.25	0.1864406779661017	0.2017362561791272	-0.0004711161687361
3	0.1864406779661017	0.2017362561791272	0.2016398528913041	-8.64229303898512 × 10 ⁻⁶
4	0.2017362561791272	0.2016398528913041	0.2016396757212823	1.035271868232712 × 10 ⁻⁶
5	0.2016398528913041	0.2016396757212823	0.2016396757234047	-2.220446049250313 × 10 ⁻¹⁶
6	0.2016396757212823	0.2016396757234047	0.2016396757234046	1.110223024625157 × 10 ⁻¹⁶

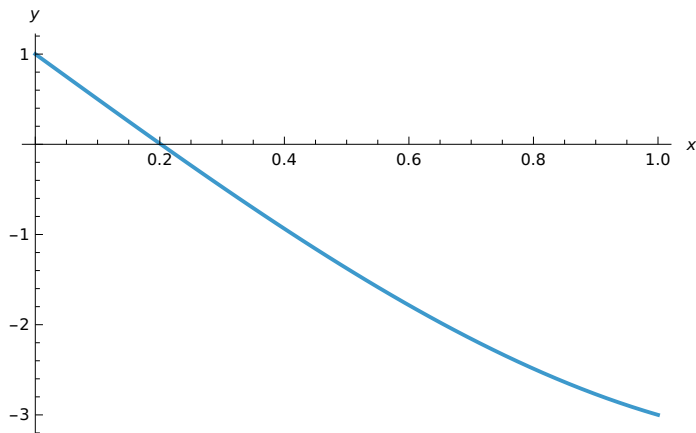
c=0.2016396757234046

```

In[26]:= NewtonRaphson[x0_, n_, f_] :=
Module[{xk1, xk = N[x0]},
k = 0;
Output = {{k, x0, f[x0]}};
While[k < n, fPrimexk = f'[xk]; If[fPrimexk == 0,
Print["The Derivative of Function at", k,
"th iteration is Zero, we cannot proceed further with the iterative scheme"];
Break[]];
xk1 = xk - f[xk] / fPrimexk;
xk = xk1;
k++;
Output = Append[Output, {k, xk, f[xk]}];
Print[NumberForm[TableForm[Output, TableHeadings → {None, {"k", "xk", "f[xk]"}}, 10]], 10];
Print["Root After", n, "iterations xk=", NumberForm[xk, 10]];
Print["Function value at approximated root, f[xk]=", NumberForm[f[xk], 6]];
f[x_] := x^3 - 5*x + 1
Plot[f[x], {x, 0, 1}, PlotStyle → {Thick}, AxesLabel → {x, y}]
Print["y=f[x]=", f[x]];
NewtonRaphson[0.5, 5, f]

```

Out[28]=



$$y=f[x]=1-5x+x^3$$

k	xk	f[xk]
0	0.5	-1.375
1	0.1764705882	0.1231426827
2	0.2015680743	0.0003492763989
3	0.2016396751	$3.100484314 \times 10^{-9}$
4	0.2016396757	$1.110223025 \times 10^{-16}$
5	0.2016396757	$1.110223025 \times 10^{-16}$

Root After 5 iterations xk=0.2016396757

Function value at approximated root, f[xk]= 1.11022×10^{-16}

```

In[1]:= A = {{1, 1, -1}, {2, 3, 5}, {3, 2, -3}};
b = {2, -3, 6};
n = Length[A];
U = A;
L = IdentityMatrix[n];
For[p = 1, p ≤ n - 1, p = p + 1, {
For[i = p + 1, i ≤ n, i = i + 1, {
m = A[[i, p]] / A[[p, p]];
L[[i, p]] = m;
A[[i]] = A[[i]] - m A[[p]];
U = A;}}];];];
Print["L=", MatrixForm[L]];
Print["U=", MatrixForm[U]];
(*Forward Substitution to solve Ly=b*)
y = ConstantArray[0, n];
y[[1]] = b[[1]] / L[[1, 1]];
For[i = 2, i ≤ n, i++,
y[[i]] = (b[[i]] - Sum[L[[i, j]] * y[[j]], {j, 1, i - 1})) / L[[i, i]];
(*BackwardSubstitution to solve Ux=y*)
x = ConstantArray[0, n];
x[[n]] = y[[n]] / U[[n, n]];
For[i = n - 1, i ≥ 1, i = i - 1,
x[[i]] = (y[[i]] - Sum[U[[i, j]] * x[[j]], {j, i + 1, n})) / U[[i, i]];
Print["Solution of the givensystem oflinear equations:{x1,x2,x3}=", x];

```

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & -1 & 1 \end{pmatrix}$$

$$U = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 7 \\ 0 & 0 & 7 \end{pmatrix}$$

Solution of the givensystem oflinear equations:{x1,x2,x3}={1, 0, -1}

```

In[41]:= Jacobi[A0_, B0_, X0_, max_] :=
Module[{A = N[A0], B = N[B0], i, j, k = 0, n = Length[X0], X = X0, Xold = X0},
Print["X", 0, "=", X];
While[k < max, For[i = 1, i ≤ n, i = i + 1,
X[[i]] = (B[[i]] + A[[i, i]*Xold[[i]] -
Sum[A[[i, j]*Xold[[j]], {j, i, n}])/A[[i, i]];
Print["X", k + 1, "=", X];
Xold = X;
k = k + 1;]];
A0 = {{4, 1, 0, 1}, {1, 4, 1, 0}, {0, 1, 4, 1}, {1, 0, 1, 4}};
B0 = {2, -2, 2, -2};
X0 = {0, 0, 0, 0};
Jacobi[A0, B0, X0, 20];
X0={0, 0, 0, 0}
X1={0.5, -0.5, 0.5, -0.5}
X2={0.75, -0.625, 0.625, -0.5}
X3={0.78125, -0.65625, 0.625, -0.5}
X4={0.789063, -0.65625, 0.625, -0.5}
X5={0.789063, -0.65625, 0.625, -0.5}
X6={0.789063, -0.65625, 0.625, -0.5}
X7={0.789063, -0.65625, 0.625, -0.5}
X8={0.789063, -0.65625, 0.625, -0.5}
X9={0.789063, -0.65625, 0.625, -0.5}
X10={0.789063, -0.65625, 0.625, -0.5}
X11={0.789063, -0.65625, 0.625, -0.5}
X12={0.789063, -0.65625, 0.625, -0.5}
X13={0.789063, -0.65625, 0.625, -0.5}
X14={0.789063, -0.65625, 0.625, -0.5}
X15={0.789063, -0.65625, 0.625, -0.5}
X16={0.789063, -0.65625, 0.625, -0.5}
X17={0.789063, -0.65625, 0.625, -0.5}
X18={0.789063, -0.65625, 0.625, -0.5}
X19={0.789063, -0.65625, 0.625, -0.5}
X20={0.789063, -0.65625, 0.625, -0.5}

```

```

In[51]:= Jacobi[A0_, B0_, X0_, max_] :=
Module[{A = N[A0], B = N[B0], i, j, k = 0, n = Length[X0], X = X0, Xold = X0},
Print["X", 0, "=", X];
While[k < max, For[i = 1, i ≤ n, i = i + 1,
X[[i]] = (B[[i]] -
Sum[A[[i, j]]*X[[j]], {j, 1, i - 1}] -
Sum[A[[i, j]]*Xold[[j]], {j, 1 + i, n}))/A[[i, i]]];
Print["X", k + 1, "=", NumberForm[X, 10]];
Xold = X;
k = k + 1;]];
A0 = {{2, -1, 0, 0}, {-1, 2, -1, 0}, {0, -1, 2, -1}, {0, 0, -1, 2}};
B0 = {1, 0, 0, 1};
X0 = {0.5, 0.5, 0.5, 0.5};
Jacobi[A0, B0, X0, 10]
X0={0.5, 0.5, 0.5, 0.5}
X1={0.75, 0.625, 0.5625, 0.78125}
X2={0.8125, 0.6875, 0.734375, 0.8671875}
X3={0.84375, 0.7890625, 0.828125, 0.9140625}
X4={0.89453125, 0.861328125, 0.8876953125, 0.9438476563}
X5={0.9306640625, 0.9091796875, 0.9265136719, 0.9632568359}
X6={0.9545898438, 0.9405517578, 0.9519042969, 0.9759521484}
X7={0.9702758789, 0.9610900879, 0.9685211182, 0.9842605591}
X8={0.9805450439, 0.9745330811, 0.9793968201, 0.98969841}
X9={0.9872665405, 0.9833316803, 0.9865150452, 0.9932575226}
X10={0.9916658401, 0.9890904427, 0.9911739826, 0.9955869913}

```

```

In[8]:= lagrange[x0_, f0_] :=
Module[{xi = x0, fi = f0, n, m, polynomial},
n = Length[xi];
m = Length[fi];
For[i = 1, i ≤ n, i++,
L[i, x_] = Product[(x - xi[[j]])/(xi[[i]] - xi[[j]]), {j, 1, i - 1}] *
Product[(x - xi[[j]])/(xi[[i]] - xi[[j]]), {j, i + 1, n}];];
polynomial[x] = Sum[L[k, x] * fi[[k]], {k, 1, n}];
Return [polynomial[x]];
]
nodes = {0, 1, 3};
values = {1, 3, 55};
lagrangepolynomial[x_] = lagrange[nodes, values]
lagrangepolynomial[x_] = Simplify[lagrangepolynomial[x]]
Print["Lagrange Polynomial=", lagrangepolynomial[x]]
lagrangepolynomial[2]

```

Out[11]=

$$\frac{1}{3} (1 - x) (3 - x) + \frac{3}{2} (3 - x) x + \frac{55}{6} (-1 + x) x$$

Out[12]=

$$1 - 6 x + 8 x^2$$

Lagrange Polynomial=1 - 6 x + 8 x²

Out[14]=

21


```

In[25]:= sum = 0;
points = {{3, 293}, {5, 508}, {6, 585}, {9, 764}};
n = Length[points]
y = points[[All, 1]]
f = points[[All, 2]]
dd[k_] := Sum[(f[[i]] / Product[If[Equal[j, i], 1, (y[[i]] - y[[j]])], {j, 1, k}]), {i, 1, k}]
p[x_] = Sum[(dd[i] * Product[If[i ≤ j, 1, x - y[[j]]], {j, 1, i - 1}]), {i, 1, n}]
Simplify[p[x]]
Evaluate[p[2.5]]

```

Out[27]=

4

Out[28]=

{3, 5, 6, 9}

Out[29]=

{293, 508, 585, 764}

Out[31]=

$$293 + \frac{215}{2} (-3 + x) - \frac{61}{6} (-5 + x) (-3 + x) + \frac{35}{36} (-6 + x) (-5 + x) (-3 + x)$$

Out[32]=

$$\frac{1}{36} (-9702 + 9003 x - 856 x^2 + 35 x^3)$$

Out[33]=

222.288

```

In[1]:= f[x_] := 1/(1 + x);
        Trap[a_, b_, n_] :=
        Module[{h},
        h = (b - a)/n;
        (h/2)*(f[a] + f[b]) + h*Sum[f[a + k h], {k, 1, n - 1}]
        N[Trap[0, 1, 2], 15]

```

Out[3]= 0.7083333333333333

```

In[101]:= f[x_] := 1/(1 + x);
           Trap[a_, b_, n_] :=
           Module[{h},
           h = (b - a)/n;
           (h/2)*(f[a] + f[b]) + h*Sum[f[a + k h], {k, 1, n - 1}]
           N[Trap[0, 1, 5], 15]

```

Out[103]= 0.695634920634921

 **SetDelayed:** Tag List in {293, 508, 585, 764}[x_] is Protected.

Out[65]=

```
0.166667 {293., 508., 585., 764.}[0.5] +  
0.333333 ((293., 508., 585., 764.)[0.25] + {293., 508., 585., 764.}[0.75]) +  
0.0833333 ((293., 508., 585., 764.)[0.] + {293., 508., 585., 764.}[1.])
```

 **SetDelayed:** Tag List in {293, 508, 585, 764}[x_] is Protected.

Out[68]=

```
0.166667 {293., 508., 585., 764.}[0.5] +  
0.333333 ((293., 508., 585., 764.)[0.25] + {293., 508., 585., 764.}[0.75]) +  
0.0833333 ((293., 508., 585., 764.)[0.] + {293., 508., 585., 764.}[1.])
```

 **SetDelayed:** Tag List in {293, 508, 585, 764}[x_] is Protected.

Out[77]=

```
0.166667 {293., 508., 585., 764.}[0.5] +  
0.333333 ((293., 508., 585., 764.)[0.25] + {293., 508., 585., 764.}[0.75]) +  
0.0833333 ((293., 508., 585., 764.)[0.] + {293., 508., 585., 764.}[1.])
```

Out[78]=

ClearAll

 **SetDelayed:** Tag List in {293, 508, 585, 764}[x_] is Protected.

Out[81]=

```
0.166667 {293., 508., 585., 764.}[0.5] +  
0.333333 ((293., 508., 585., 764.)[0.25] + {293., 508., 585., 764.}[0.75]) +  
0.0833333 ((293., 508., 585., 764.)[0.] + {293., 508., 585., 764.}[1.])
```

Out[85]=

0.693254

```
In[86]:= ClearAll[Simp, f];
f[x_] := 1/(1 + x);
Simp[a_, b_, n_] :=
Module[{h},
h = (b - a)/(2 n);
(h/3)*(f[a] + f[b] +
4*Sum[f[a + (2 k - 1) h], {k, 1, n}] +
2*Sum[f[a + 2 k h], {k, 1, n - 1}])]
N[Simp[0, 1, 2], 10]
```

```
Out[89]=
0.6932539683
```

```
In[94]:= ClearAll[Simp, f];
f[x_] := 1/(1 + x);
Simp[a_, b_, n_] :=
Module[{h},
h = (b - a)/(2 n);
(h/3)*(f[a] + f[b] +
4*Sum[f[a + (2 k - 1) h], {k, 1, n}] +
2*Sum[f[a + 2 k h], {k, 1, n - 1}])]
N[Simp[0, 1, 4], 9]
```

```
Out[97]=
0.693154531
```

In[104]:=

```
Euler[{t_, y_}] := {t + h, y + h * f[t, y]}
f[t_, y_] := 1 + (y / t)
h = 0.5
points = NestList[Euler, {1, 1}, 10]
TableForm[points, TableHeadings -> {Automatic, {"t_i", "y_i"}}]
```

Out[106]=

0.5

Out[107]=

```
{{1, 1}, {1.5, 2.}, {2., 3.16667}, {2.5, 4.45833}, {3., 5.85}, {3.5, 7.325},
{4., 8.87143}, {4.5, 10.4804}, {5., 12.1448}, {5.5, 13.8593}, {6., 15.6193}}
```

Out[108]//TableForm=

	t_i	y_i
1	1	1
2	1.5	2.
3	2.	3.16667
4	2.5	4.45833
5	3.	5.85
6	3.5	7.325
7	4.	8.87143
8	4.5	10.4804
9	5.	12.1448
10	5.5	13.8593
11	6.	15.6193

```

In[8]:= RK4[a0_, b0_, e_, m0_] :=
Module[{a = a0, b = b0, j, m = m0},
h = (b - a) / m;
Y = T = Table[0, {m + 1}];
T[[1]] = a;
Y[[1]] = e;
For[j = 1, j ≤ m, j++,
K1 = h f[T[[j]], Y[[j]]];
K2 = h f[T[[j]] + (h / 2), Y[[j]] + (K1 / 2)];
K3 = h f[T[[j]] + (h / 2), Y[[j]] + (K2 / 2)];
K4 = h f[T[[j]] + h, Y[[j]] + K3];
Y[[j + 1]] = Y[[j]] + (1 / 6) (K1 + K2 + K3 + K4);
T[[j + 1]] = a + h j;];
Return[Transpose[{T, Y}]]

f[t_, y_] := t^2 - y;
points = RK4[0, 0.8, 1, 8];
TableForm[points,
TableHeadings → {Automatic, {"t_i", "y_i"}}]

```

Out[11]//TableForm=

	t_i	y_i
1	0	1
2	0.1	0.93679
3	0.2	0.878868
4	0.3	0.827167
5	0.4	0.782562
6	0.5	0.745872
7	0.6	0.717863
8	0.7	0.699254
9	0.8	0.690718